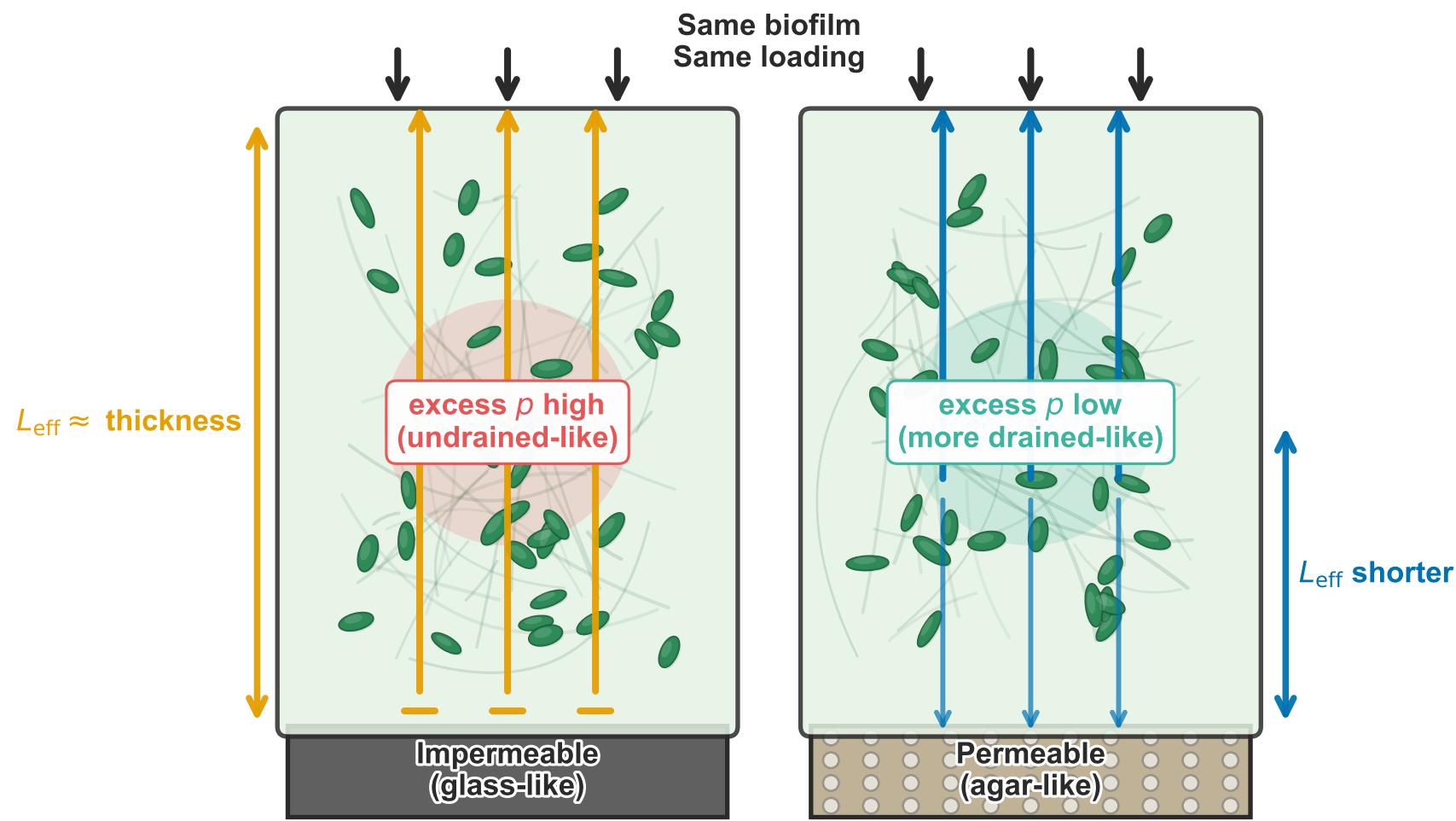
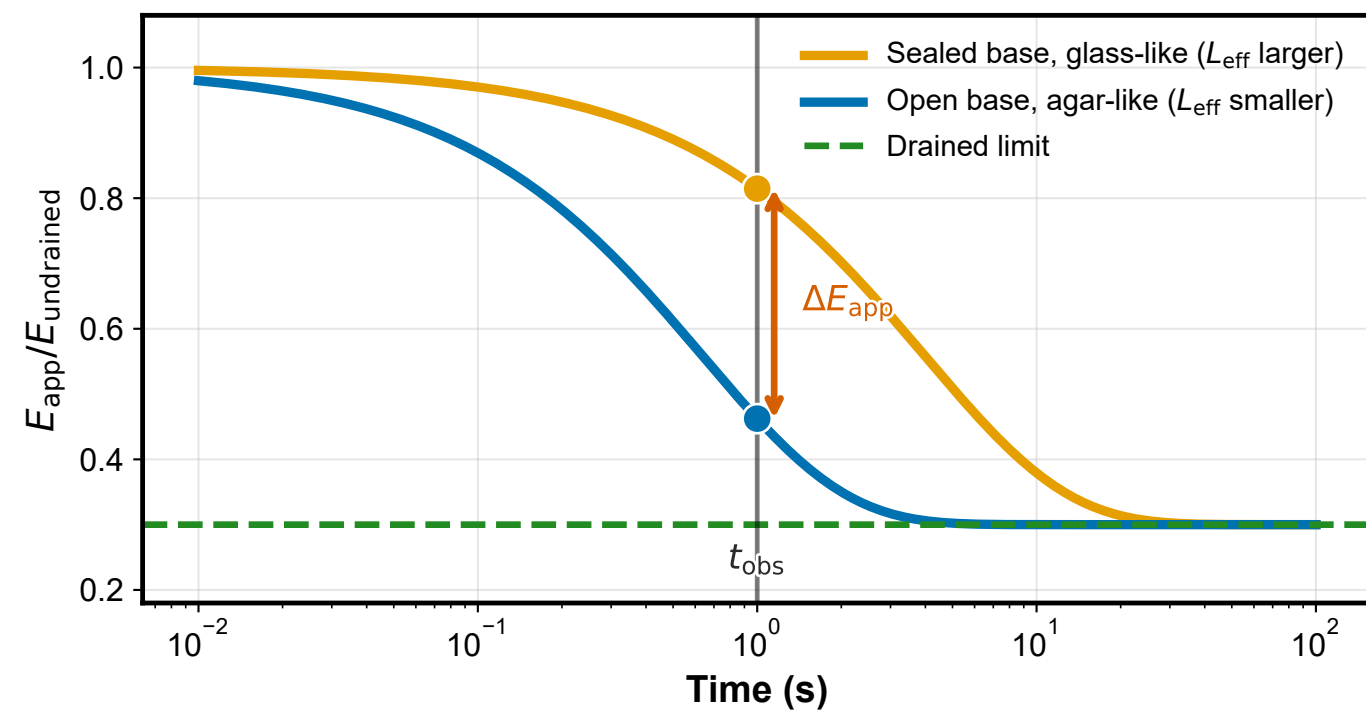


(a) Drainage boundary sets effective path length L_{eff}



$\tau_{\text{drain}} \propto L_{\text{eff}}^2$
 Boundary sensitivity is largest when $t_{\text{obs}} \sim \tau_{\text{drain}}$
 At matched t_{obs} , larger L_{eff} yields higher E_{app}

(b) Readout-time dependence at fixed t_{obs}



(c) Boundary perturbation diagnostic $E_{\text{app}}(k_b)$

